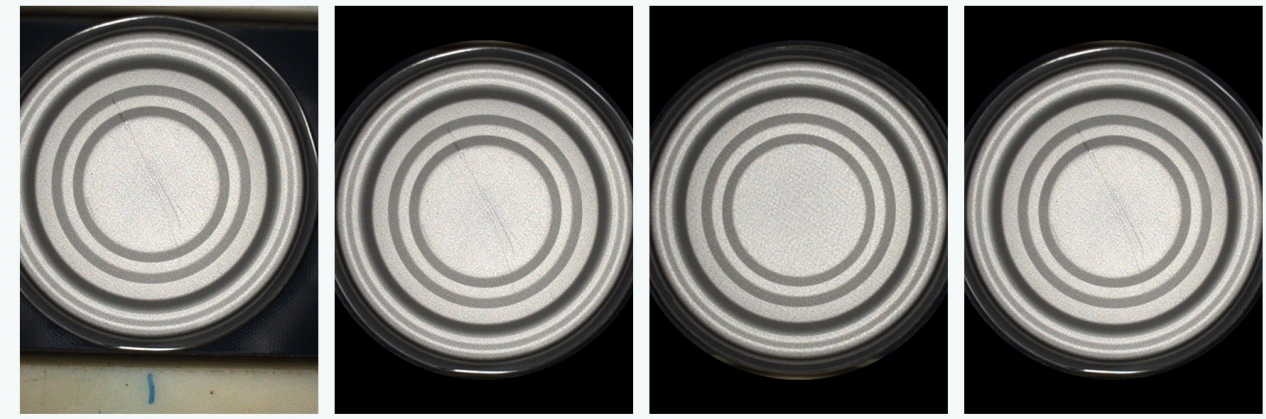
TECHNICAL PROPOSAL / PROOF OF CONCEPT

# AI-Based Visual Quality Inspection for Tomato Paste Can Lids

A focused demonstration and phased development proposal




Prepared for

## Rojin Taak Agro Industries

October 2026 | Demonstration based on a public lid-image dataset

**Scope statement** This is an offline, lid-only feasibility demonstration. Validation on Rojin product images and an on-site production pilot are proposed next steps.

01 / EXECUTIVE OVERVIEW

# Executive Summary

A measured demonstration with a deliberately limited scope.

This proposal presents an AI-assisted vision prototype for classifying tomato paste can lid images into acceptable and defective categories. It uses a pretrained YOLO11n classification model fine-tuned on lid images. A separate decision layer translates model scores into three operational actions: GOOD, DEFECTIVE, and REVIEW.

The demonstration is designed to provide an objective basis for considering a Rojin-specific trial, not to claim readiness for a complete factory inspection installation. The 1,307-image dataset used for this development is public/external, not a dataset collected from Rojin production.

## Measured results: offline hold-out evaluation

| Indicator | Observed result | Interpretation |
|---|---|---|
| **Test set** | 261 masked images | 114 acceptable; 147 defective |
| **Correct operational decisions** | 257 / 261 (98.47%) | REVIEW is counted as not automatically correct |
| **Reviewed by operator** | 3 / 261 (1.15%) | All three review samples labeled good |
| **Defective lids released as GOOD** | 0 / 147 | No observed false accept in this small external-data split |
| **Processing throughput** | 62.36 images/s | Batched GPU test on already-masked image files |

## Proposed value to Rojin

- Demonstrate the feasibility of consistent, probability-based lid inspection before committing to line integration.
- Use an explicit REVIEW path so borderline decisions can be routed to a quality operator.
- Establish a scalable technical foundation for future body, label and additional packaging inspections, subject to new data and engineering work.

**Decision requested** Review this proof of concept and authorize a factory-specific image collection and feasibility pilot before any production-level performance commitment.

02 / BUSINESS CONTEXT

# Inspection Objective & Project Boundaries

Lid quality is the only inspection target of the current demo.

## Current demonstration

The prototype classifies visual patterns within a segmented top-view image of a metal can lid. The intended inspection subject is the tomato paste can lid; model training and tests were performed on an external lid-image dataset with visually similar, but not verified Rojin-specific, examples.

| Included now | Not yet implemented |
|---|---|
| **Top-view lid image classification** | Can sidewall and body defects |
| **GOOD / DEFECTIVE / REVIEW decision logic** | Product label presence, placement and print quality |
| **Per-image P(good) and P(defective)** | Other packaging parts, seal integrity and content quality |
| **Batch processing and exportable CSV results** | Live camera acquisition, PLC integration and rejection mechanism |
| **Masked-image pre-processing workflow** | Validated inspection of any other Rojin product |

## Factory information needed for engineering

The later industrial phase requires confirmation of the actual lid geometries, defect taxonomy, camera location, illumination, conveyor speed and spacing, environmental constraints, reject mechanism and acceptance criteria. These have not yet been established from Rojin site measurements.

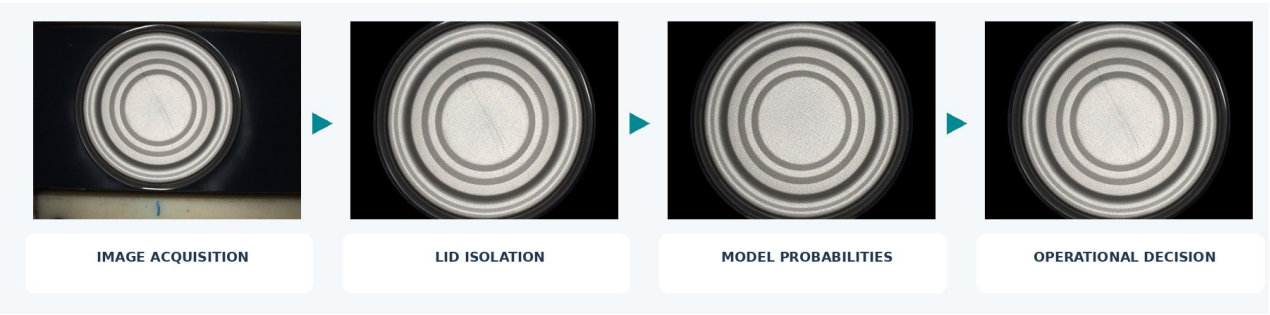
The ibea proposal provided as a reference distinguishes product specifications, line specifications, engineering constraints and rejection stations. That structure has been adapted here without reusing third-party equipment claims.

**Important distinction** Visible resemblance between public-dataset lids and Rojin tomato paste lids supports feasibility exploration; it does not establish field accuracy or product compatibility.

03 / SYSTEM ARCHITECTURE

# Proposed Inspection Workflow

Current offline architecture and its future industrial interpretation.




*Illustrative sequence using sample images from the public dataset. The camera acquisition shown is a proposed input stage, not an installed Rojin camera.*

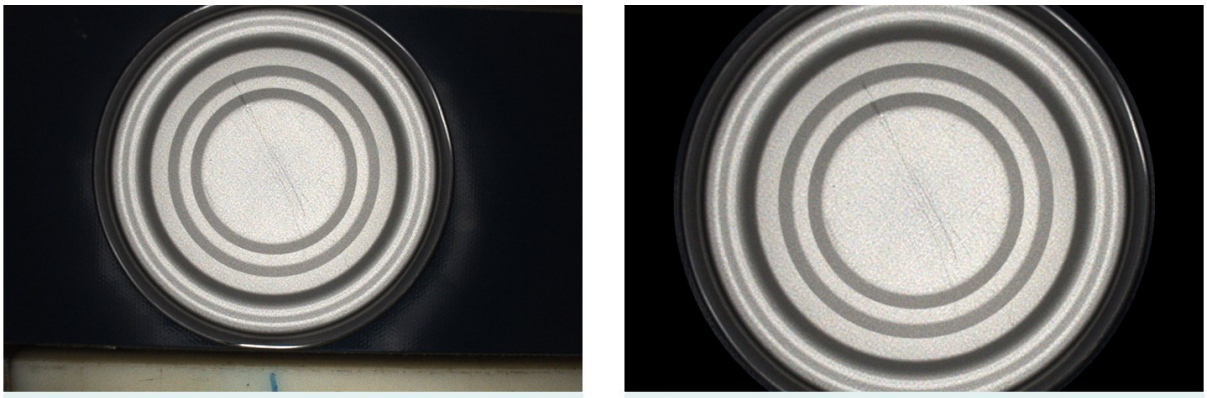
## Implemented software workflow

1. Raw images are supplied from file storage. An OpenCV-based geometric routine estimates the circular lid region and produces a dark-background masked image. The mask-generation workflow includes visual audit outputs and a manual override path.

2. YOLO11n-cls estimates the probability of the two trained labels: good and defective. The network is a classification model; it does not provide pixel-level defect segmentation or identify the physical dimensions of a flaw.

3. A post-classification decision rule assigns GOOD, DEFECTIVE or REVIEW. A batch evaluation script records per-image scores, ground-truth folder labels and final decisions.

## Original input versus segmented image




*Actual raw and masked sample included in the project archive. Geometric lid isolation precedes classification.*

**Deployment boundary** A production pipeline still needs camera triggering, lighting, synchronization, streaming segmentation, PLC/reject signaling, fail-safe handling and on-site validation.

04 / DATASET EVIDENCE

# Dataset & Training Design

Public image data, separate development and testing sets.

The archived public TinplateLidDATA2024 image collection contains 1,307 labeled examples: 571 good and 736 defective. Masked variants were generated, then the images were assigned to a stratified, seeded 80/20 development/test split. The development pool was further partitioned into training and validation subsets.




*Image-level split; test membership was preserved outside the model-training directory.*

| Subset | Good | Defective | Total |
|---|---|---|---|
| **Training** | 412 | 531 | 943 |
| **Validation** | 45 | 58 | 103 |
| **Independent held-out test** | 114 | 147 | 261 |
| **TOTAL** | 571 | 736 | 1,307 |

## Model training configuration

| Parameter | Recorded configuration |
|---|---|
| **Architecture** | YOLO11n-cls (pretrained) |
| **Image size / batch** | 640 px / 32 |
| **Optimizer / initial learning rate** | AdamW / 0.001 |
| **Training plan / early stopping** | Up to 100 epochs / patience 20 |
| **Completed training / best epoch** | 61 epochs / epoch 41 |
| **Compute** | NVIDIA GeForce RTX 5060 GPU; mixed precision |

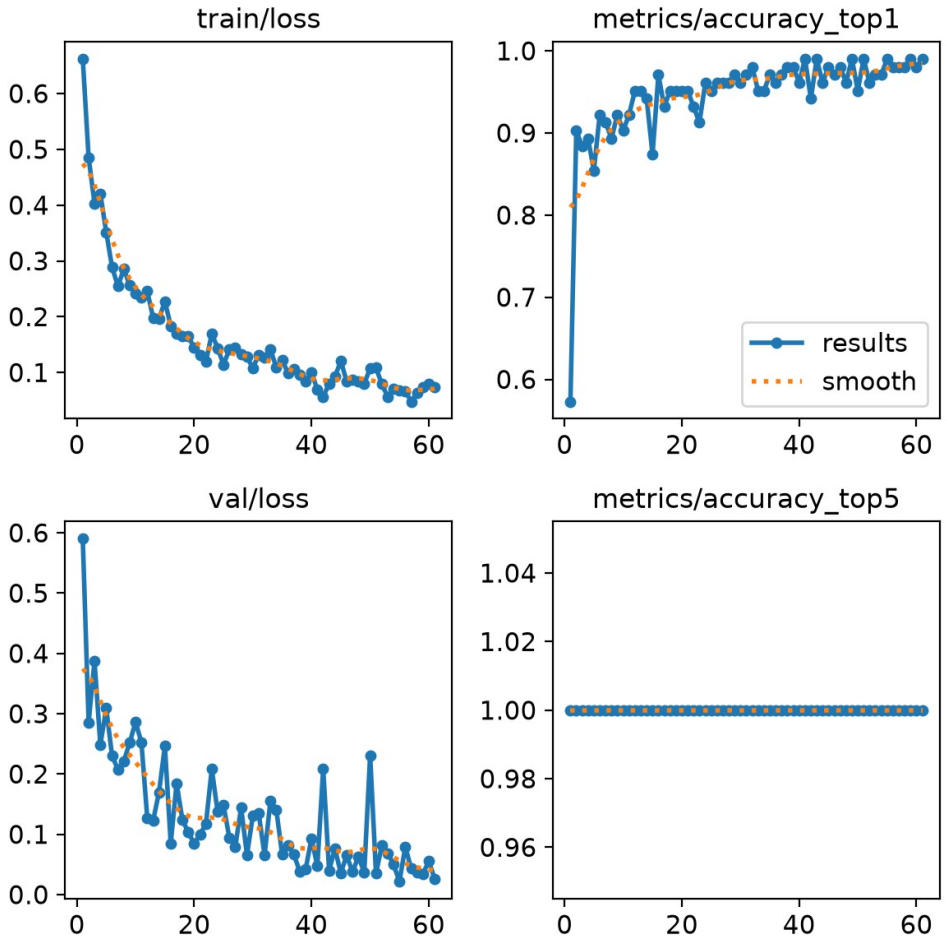
The supplied integrity audit shows exact file/content consistency across the data partitions and no identical SHA-256 files across train/test. Because the split is image-level, visually near-duplicate frames or images from related recording sessions have not been excluded by this check.

**Quality note** Automated mask-processing success does not by itself certify the visual completeness of all lid edges. Production qualification requires a targeted audit of edge defects and difficult cases.

05 / MODEL LEARNING

# Training & Validation Evidence

Saved best checkpoint: yolo11n_cls_masked_random80_v2/weights/best.pt




*Original Ultralytics training charts: training/validation loss and classification accuracy over 61 epochs.*

Training ended through Early Stopping after 61 epochs; the saved best checkpoint was obtained at epoch 41. The final best-checkpoint validation top-1 accuracy is approximately 99.03% (102 correct predictions from 103 validation images). This figure describes validation performance on held-out development data, not field performance.

## Why these graphs matter

- Learning curves show an overall reduction in classification loss as the model adapts to the masked-lid images.
- The best weights were selected using validation feedback; the final 261-image test set was kept separate from this model-selection process.
- With only two classes, top-5 accuracy is not an informative quality metric; top-1 accuracy and defect-specific errors are more relevant.

**Training caveat** The recorded run used automatic augmentation, random rescaling and image erasing. Future Rojin-specific qualification should verify that these transformations do not conceal small edge defects.

06 / DECISION POLICY

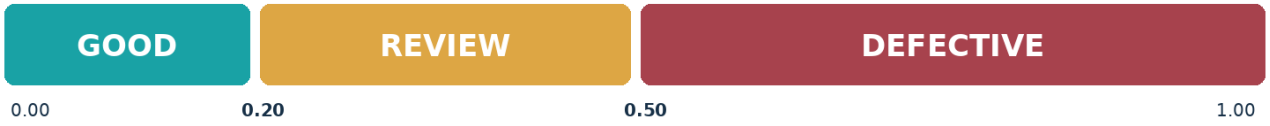
# Operational Decision Logic

Probability score bands are applied outside the trained two-class network.

The model emits P(good) and P(defective). The downstream decision module uses the defective score to determine the next action. REVIEW is an operational abstention state, not a third class learned by YOLO.

| Model output | Action | Intended handling |
|---|---|---|
| **P(defective) < 0.20** | GOOD | Candidate for acceptance under the inspected lid criterion |
| **0.20 <= P(defective) <= 0.50** | REVIEW | Hold for operator evaluation |
| **P(defective) > 0.50** | DEFECTIVE | Candidate for quality rejection |

## Decision-policy diagram




These thresholds are the values used in the supplied final test script. They are configuration choices, not calibrated defect probabilities. Before Rojin deployment, thresholds should be selected against jointly agreed false-accept and operator-review targets using Rojin-specific validation data.

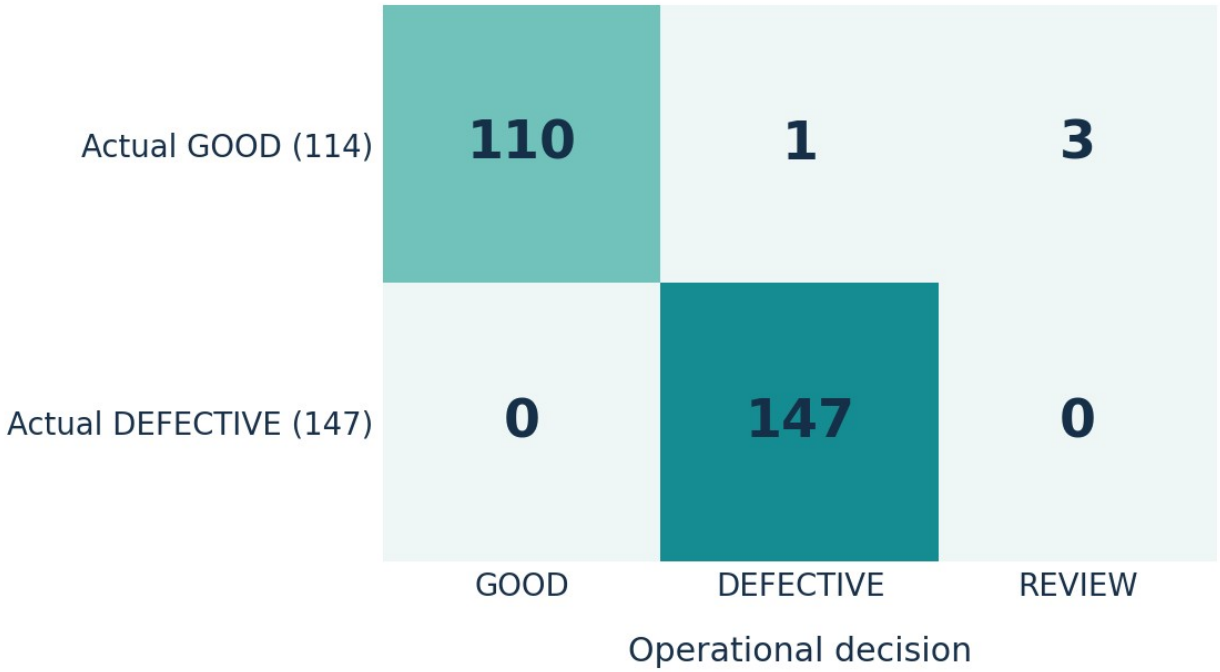
## Safety-oriented operating behavior

For a future installation, REVIEW decisions, invalid images, missing lids, camera faults and communications failures should trigger a configurable fail-safe handling path. No physical reject station has been implemented or tested in this offline demonstration.

07 / PERFORMANCE ASSESSMENT

# Held-Out Test Results

All figures on this page are calculated from the supplied 261-row predictions.csv.




*Operational confusion matrix: actual dataset class versus software decision.*

| Metric | Measured value | Definition |
|---|---|---|
| **Correct-decision rate** | 98.47% (257/261) | REVIEW counts as not automatically correct |
| **Accuracy on non-REVIEW decisions** | 99.61% (257/258) | Correct among 258 decisive outputs |
| **Defective-lid capture** | 100% (147/147) | All 147 labeled defective samples flagged DEFECTIVE |
| **Defective decision precision** | 99.32% (147/148) | One good lid received DEFECTIVE decision |
| **Operator review** | 1.15% (3/261) | Three labeled-good images routed to REVIEW |
| **False accept / false reject** | 0 / 1 | Actual defective -> GOOD / actual good -> DEFECTIVE |

This is a promising offline feasibility result, but is based on a relatively small public-dataset, image-level hold-out test. Zero observed false accepts does not imply a guaranteed zero false-accept rate in production.

**Metric integrity** The source sklearn report treats REVIEW as a third predicted class although the dataset has no true REVIEW class; its three-class macro averages are not used as industrial performance measures here.

08 / VISUAL AUDIT

# Representative Inspection Examples

Original masked images from the final hold-out test; not images from Rojin production.

**GOOD**



P(defect)=0.011 | P(defect)=0.011 | P(defect)=0.033

**DEFECTIVE**



P(defect)=1.000 | P(defect)=1.000 | P(defect)=1.000

**REVIEW**



P(defect)=0.210 | P(defect)=0.373 | P(defect)=0.216

*Rows: GOOD, DEFECTIVE, REVIEW. Each caption reports the saved P(defective) from the final test CSV.*

The REVIEW examples in this public-dataset test are all labeled good; their defective scores fall in the 0.20-0.50 interval. The one observed false rejection is a different good image, which should be examined during error analysis before any site pilot.

For the next demonstration revision, we recommend showing the original image, masked lid, model scores and human-verified label side by side, especially for borderline defects at the lid rim.

09 / CAPACITY AND EQUIPMENT

# Processing Speed & Industrial Requirements

The measured benchmark is not an end-to-end factory-line capacity claim.

## Current laboratory throughput

| Benchmark item | Recorded value |
|---|---|
| **Number of images** | 261 pre-masked PNG files |
| **Wall-clock section measured** | 4.185 s |
| **Reported throughput** | 62.36 images/s |
| **Inference method** | Batched Ultralytics prediction, batch size 32, GPU, FP16 argument |
| **Included in timer** | Prediction requests and copying test images to result folders |
| **Not included in timer** | Camera acquisition, raw-image lid segmentation, synchronization, CSV/report creation and line reject actuation |

The measured rate exceeds the initial experimental target of 25 images/s for this reduced workflow. The feasibility of at least 25 fully processed lids per second, including acquisition and segmentation, remains unverified and should be explicitly benchmarked at the factory.

## Equipment and integration to be defined for pilot

- Industrial camera, appropriate lens and illumination selected using actual Rojin lid samples.
- Mounting geometry, trigger sensor and conveyor timing to ensure a full and stable lid view.
- Industrial PC/GPU sizing, storage, operator screen and alarm/review workflow.
- PLC signaling, reject mechanism or inspection-only mode, with fail-safe behavior and traceability.

**Commercial boundary** Equipment quantities, installation costs, inspection guarantees, delivery dates and service terms require a separate site survey and commercial agreement.

10 / PROPOSED VALIDATION

# Factory Pilot & Acceptance Approach

A gated path from laboratory demonstration to site-specific evidence.

| Step | Activity | Deliverable / approval gate |
|---|---|---|
| **1. Site requirements** | Observe conveyor, measure throughput, review defects and operating constraints. | Agreed inspection scope and test plan |
| **2. Representative Rojin data** | Capture labeled good and defective lids over multiple runs, dates and lighting conditions. | Reviewed production sample set |
| **3. Model adaptation** | Recheck masks, optimize imaging, retrain or fine-tune and calibrate decision thresholds. | Rojin-specific candidate model |
| **4. Locked evaluation** | Evaluate on unseen production batches; report false accepts, false rejects, REVIEW and confidence. | Signed technical validation report |
| **5. End-to-end pilot** | Measure acquisition-to-decision throughput and operational robustness with relevant equipment. | Joint go / no-go decision for line integration |

## Suggested acceptance measures (to be agreed)

- Maximum allowed false accept rate for critical lid defects, defined using Rojin quality policy.
- False reject cost, maximum review workload and label agreement between inspectors.
- Required line throughput and inspection latency including all processing stages.
- Performance stability across batches, products, operators, illumination and production shifts.

Acceptance thresholds and test duration must be set with Rojin before on-site testing. This document intentionally does not offer numerical guarantees for unmeasured factory conditions.

11 / FUTURE ROADMAP

# Phased Expansion & Next Steps

Additional features follow only after a successful lid demonstration and scope agreement.

| Phase | Scope | Status |
|---|---|---|
| **A. Current demonstration** | Offline classification of tomato paste lid images; probability scores and review decision. | Implemented; public-data tested |
| **B. Rojin-specific validation** | Factory samples, imaging adaptation, tuned model and agreed acceptance metrics. | Proposed / not started |
| **C. Lid inspection pilot** | Camera/lighting integration, operator review and optional PLC/reject output. | Future engineering |
| **D. Expanded tomato paste QA** | Can body, label presence/position/print, further packaging checks as required. | Separate future modules |
| **E. Other Rojin products** | Model/data adaptation for different packages and product types. | Potential future scope |

## Requested next decision

Rojin is invited to review the offline demonstration and determine whether a short engineering assessment and controlled factory-data pilot should proceed. The parties would then agree on sample collection, defect definitions, integration responsibilities, acceptance metrics, commercial scope and schedule.

## Document basis and references

Primary project evidence: supplied Rojin AI project review archive (training configuration, result curves, final-test predictions, and image samples), October 2026.

Reference documents supplied for proposal structuring and industrial context: (1) ibea GmbH, “The Cube - Visual Inspection System for Cans”, proposal P70-Rojin-2018-01, 9 October 2018 (especially pp. 3-8, 12); (2) Parmacontrols, “Canning & Filling HiControls” illustrated brochure (pp. 1-4); (3) Shanghai UPG, “Metal Printing Offline Inspection” technical proposal (especially pp. 3-9). Third-party performance specifications are not presented as performance of this prototype.

**Proposal status** Technical discussion draft. The current result is an offline proof of concept using a public dataset. Product-specific accuracy, throughput and commercial terms remain subject to validation and agreement.